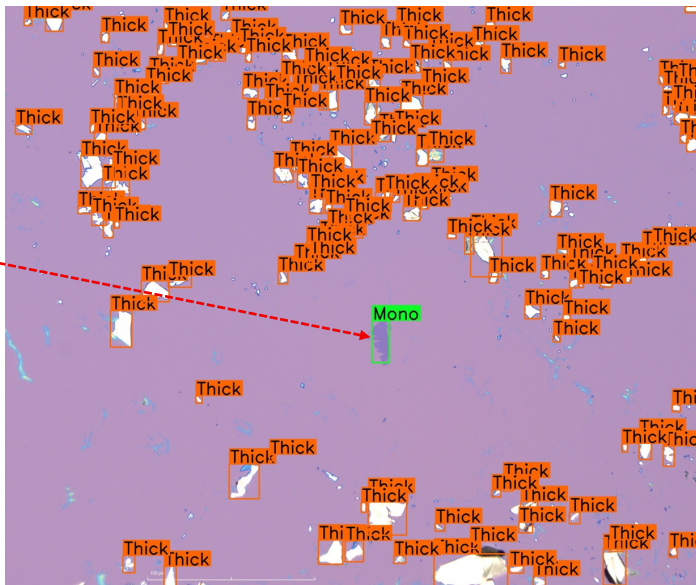
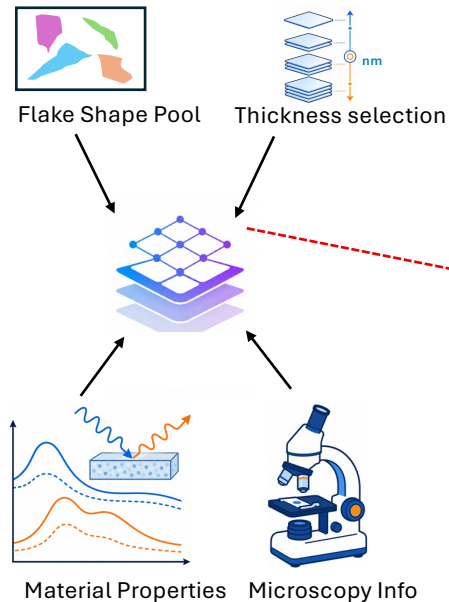


Synthia

QMat-Instruct Dataset

QuPAINT: Physics-Aware Instruction Tuning



How many monolayer flakes are there in the image?



I first scan the image and mark all flake-like regions, ordering them from top-left to bottom-right as <boxes>. Most of these candidates are either too dark or show clear, high-contrast edges, which is more consistent with thicker flakes. One region at <boxes> appears extremely faint with only a slight hue shift relative to the purple SiO_2/Si background. Its boundaries fade gradually into the substrate, giving a semi-transparent appearance rather than an opaque outline. These visual cues—very low contrast, subtle color shift, and soft edges—are characteristic of a monolayer. Conclusion: There is 1 monolayer flake at <boxes>.

Confirm if this sample includes mono layer flakes.



I review all candidate flakes at <boxes> and evaluate each against the expected appearance of a monolayer. A true monolayer should remain close to the background intensity, with only subtle optical contrast and a gentle color shift, while its edges should appear semi-transparent rather than abrupt. Among the candidates, the flake at <boxes> matches these criteria well: it is very faint, low contrast, and softly blended with the substrate, while the others are darker or exhibit stronger boundaries. Conclusion: Yes, this sample contains a monolayer flake.

Locate all the mono layer flakes in the image.



I identify all flake-like regions and then narrow them down to the monolayer candidates. Inspecting each region, only the area at <boxes> shows the low-contrast, faint appearance with a subtle hue shift and semi-transparent edges characteristic of a monolayer. The remaining candidates are more saturated, higher contrast, or have crisper boundaries, indicating thicker flakes. Conclusion: The monolayer flake is located at <boxes>.